Supplementary Material

Mathematical Definitions, Notation, and Computational Procedures
Paper 6: A Conservation Constraint in LLM Context Processing
Across Four Epistemological Domains

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1 Experimental Conditions

Each model-domain run consists of three conditions, each comprising 50 independent trials of a 30-prompt conversation.

TRUE condition Each trial presents the model with a coherent 29-message conversational history followed by a prompt. The history is the same across all 50 trials within a run. Context is semantically and temporally coherent.

COLD condition Each trial presents the model with the same prompt in isolation, with no prior conversational history. This is the context-free baseline.

SCRAMBLED condition Each trial presents the model with the same 29 messages from the TRUE history, but in a uniformly randomised order, followed by the same prompt. Content is present; temporal order is destroyed.

All conditions use temperature = 0.7 and max_tokens = 1024. Embeddings are computed using all-MiniLM-L6-v2 (384-dimensional) as the primary model, with all-mpnet-base-v2 (768-dimensional) and LaBSE (768-dimensional) for robustness validation.

2 Core Metrics

2.1 Relational Coherence Index (RCI)

For a given condition $c \in \{\text{TRUE}, \text{COLD}, \text{SCRAMBLED}\}$, model m, domain d, and prompt position $p \in \{1, \ldots, 30\}$:

Let $\mathbf{e}_{i,p}^{(c)}$ denote the embedding of the response generated in trial i at position p under condition c.

The RCI at position p is the mean pairwise cosine similarity across $N = 50$ trials:

$$\text{RCI}_p^{(c)} = \frac{2}{N(N-1)} \sum_{i<j} \cos\left(\mathbf{e}_{i,p}^{(c)}, \mathbf{e}_{j,p}^{(c)}\right) \tag{1}$$

The model-level RCI is the mean across all 30 positions:

$$\overline{\text{RCI}}^{(c)} = \frac{1}{30} \sum_{p=1}^{30} \text{RCI}_p^{(c)} \tag{2}$$

2.2 Delta Relational Coherence Index (ΔRCI)

ΔRCI measures how much the conversational context increases response coherence relative to the context-free baseline:

$$\Delta\text{RCI} = \overline{\text{RCI}}^{(\text{TRUE})} - \overline{\text{RCI}}^{(\text{COLD})} \tag{3}$$

$\Delta\text{RCI} > 0$ indicates that context increases response alignment across trials. $\Delta\text{RCI} < 0$ indicates divergent context coupling (context *increases* response variability).

2.3 Variance Ratio (Var_Ratio)

The variance of responses at position p under condition c is computed from the pairwise cosine similarity distribution across all $N = 50$ trial pairs. Let:

$$S_p^{(c)} = \left\{\cos\left(\mathbf{e}_{i,p}^{(c)}, \mathbf{e}_{j,p}^{(c)}\right) : 1 \leq i < j \leq N\right\} \tag{4}$$

be the set of all $\binom{N}{2} = 1225$ pairwise cosine similarities at position p. The positional variance is:

$$\mathrm{Var}_p^{(c)} = \mathrm{Var}\left(S_p^{(c)}\right) \tag{5}$$

The model-level Var_Ratio is:

$$\mathrm{Var\_Ratio} = \frac{\overline{\mathrm{Var}}^{(\mathrm{TRUE})}}{\overline{\mathrm{Var}}^{(\mathrm{COLD})}} \quad \text{where} \quad \overline{\mathrm{Var}}^{(c)} = \frac{1}{30}\sum_{p=1}^{30} \mathrm{Var}_p^{(c)} \tag{6}$$

This formulation measures the spread of the pairwise similarity distribution rather than dimension-wise embedding variance, capturing how consistently the model produces similar responses across trials at each position.

Var_Ratio < 1: context reduces response variance (convergent coupling).
Var_Ratio > 1: context increases response variance (divergent coupling).

2.4 Variance Reduction Index (VRI)

$$\mathrm{VRI} = 1 - \mathrm{Var\_Ratio} \tag{7}$$

VRI is used in entanglement analysis (Section 6). Positive VRI indicates convergent coupling; negative VRI indicates divergent coupling.

3 The Conservation Product K

3.1 Definition

For a given model-domain run, the conservation product is:

$$K = \Delta\mathrm{RCI} \times \mathrm{Var\_Ratio} \tag{8}$$

3.2 Domain-Level Statistics

For domain d with N_d model-domain runs yielding values $K_1^{(d)}, K_2^{(d)}, \ldots, K_{N_d}^{(d)}$:

$$\bar{K}^{(d)} = \frac{1}{N_d}\sum_{i=1}^{N_d} K_i^{(d)} \quad \text{(domain mean)} \tag{9}$$

$$\mathrm{SD}^{(d)} = \sqrt{\frac{1}{N_d - 1}\sum_{i=1}^{N_d}\left(K_i^{(d)} - \bar{K}^{(d)}\right)^2} \quad \text{(standard deviation)} \tag{10}$$

$$\mathrm{CV}^{(d)} = \frac{\mathrm{SD}^{(d)}}{\bar{K}^{(d)}} \quad \text{(coefficient of variation)} \tag{11}$$

CV is used as the primary measure of within-domain conservation tightness. Lower CV indicates tighter clustering of K values within the domain.

3.3 Domain Values (Primary Embedding: MiniLM 384D)

Domain	N	$\bar{K}$	SD	CV
Medical (Discovered)	8	0.429	0.073	0.170
Legal (Argued)	5	0.348	0.074	0.214
Philosophy (Explored)	6	0.301	0.050	0.166
Applied Ethics (Felt)	5	0.223	0.036	0.162

Table 1: Domain-level K statistics under primary embedding (all-MiniLM-L6-v2).

4 Content-Order Decomposition

The SCRAMBLED condition separates ΔRCI into two components:

$$\Delta\mathrm{RCI}_{\mathrm{content}} = \overline{\mathrm{RCI}}^{(\mathrm{SCRAM})} - \overline{\mathrm{RCI}}^{(\mathrm{COLD})} \quad \text{(content component)} \tag{12}$$

$$\Delta\mathrm{RCI}_{\mathrm{order}} = \overline{\mathrm{RCI}}^{(\mathrm{TRUE})} - \overline{\mathrm{RCI}}^{(\mathrm{SCRAM})} \quad \text{(order component)} \tag{13}$$

The content fraction is:

$$\text{Content Fraction} = \frac{\Delta\mathrm{RCI}_{\mathrm{content}}}{\Delta\mathrm{RCI}} \times 100\% = \frac{\overline{\mathrm{RCI}}^{(\mathrm{SCRAM})} - \overline{\mathrm{RCI}}^{(\mathrm{COLD})}}{\overline{\mathrm{RCI}}^{(\mathrm{TRUE})} - \overline{\mathrm{RCI}}^{(\mathrm{COLD})}} \times 100\% \tag{14}$$

Content Fraction $= 100\%$ means SCRAMBLED $\approx$ TRUE: conversational order carries no additional information. Content Fraction $= 0\%$ means SCRAMBLED $\approx$ COLD: all context sensitivity is carried by order, not content.

5 Exploration Arc

The Exploration Arc measures whether response variance converges or diverges over the course of the 30-prompt conversation:

$$\mathrm{Arc} = \frac{\mathrm{Var}_{30}^{(\mathrm{TRUE})}}{\mathrm{Var}_{1}^{(\mathrm{TRUE})}} \tag{15}$$

Arc < 1: responses converge (become more similar across trials) as conversation deepens. This is the pattern for Medical (Discovered truth), where diagnostic conclusions narrow with accumulated context.

Arc > 1: responses diverge (become more variable across trials) as conversation deepens. This is the pattern for Philosophy (Explored truth), where inquiry expands rather than converges.

6 Entanglement

Entanglement measures the coupling between context sensitivity and variance reduction across the 30 prompt positions within a single model-domain run.

For each position p, compute position-level $\Delta\mathrm{RCI}_p$ and $\mathrm{VRI}_p = 1 - \mathrm{Var\_Ratio}_p$.

Entanglement is the Pearson correlation across all 30 positions:

$$r_{\mathrm{ent}} = \mathrm{Pearson}\left(\{\Delta\mathrm{RCI}_p\}_{p=1}^{30},\ \{\mathrm{VRI}_p\}_{p=1}^{30}\right) \tag{16}$$

Statistical significance tested at $\alpha = 0.05$ (two-sided).

Positive r_{ent}: positions where context sensitivity is high also show greater variance reduction. This is *convergent entanglement*.

Negative r_{ent}: positions where context sensitivity is high show greater variance *amplification*. This is *divergent entanglement*.

7 Effective Rank (from Paper 9)

The effective rank of an embedding matrix $\mathbf{E} \in \mathbb{R}^{N\times D}$ is computed as the exponential of the Shannon entropy of the normalised singular value spectrum:

$$\text{eff\_rank}(\mathbf{E}) = \exp(H(\mathbf{p})) \quad \text{where} \quad p_k = \frac{\sigma_k}{\sum_j \sigma_j} \tag{17}$$

and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(N,D)} \geq 0$ are the singular values of $\mathbf{E}$ and $H(\mathbf{p}) = -\sum_k p_k \log p_k$ is the Shannon entropy of the normalised spectrum.

Effective rank $\approx \min(N,D)$ indicates a full-rank, non-degenerate embedding space. Effective rank ≈ 1 indicates near-complete collapse to a single dimension. MuRIL achieves effective rank $= 1.42/10$; MiniLM achieves $9.88/10$ (Paper 9, https://doi.org/10.5281/zenodo.19466613).

8 Entanglement Stability Index (ESI, from Paper 4)

The Entanglement Stability Index predicts which models will exhibit multi-turn instability (the "Lost in Conversation" phenomenon):

$$\text{ESI} = \frac{1}{|1 - \text{Var\_Ratio}|} \tag{18}$$

ESI measures the proximity of Var_Ratio to unity. When Var_Ratio ≈ 1, context neither amplifies nor reduces variance, yielding high ESI (stable). When Var_Ratio deviates substantially from 1 (either convergent or divergent), ESI is low (unstable).

- ESI > 2.0: Stable multi-turn behaviour
- ESI < 1.0: Elevated instability risk
- ESI $\to \infty$: Var_Ratio $= 1.0$ (perfect stability)
- ESI $\to 0$: Extreme divergent or convergent coupling

Paper 4 [Laxman, 2026d] demonstrated that ESI < 1.0 at task-critical positions (Medical P30) identifies models exhibiting the "Lost in Conversation" phenomenon documented by Laban et al. (2025) across 200,000+ conversations.

Model	**Var_Ratio (P30)**	**ESI**	**Assessment**
Llama 4 Scout	7.46	0.15	Unstable
Llama 4 Maverick	2.64	0.61	Caution
Qwen3 235B	1.02	50.0	Stable
GPT-4o	0.58	2.38	Stable
Claude Haiku	0.65	2.86	Stable

Table 2: ESI at medical summarisation position (P30). From Paper 4 [Laxman, 2026d].

9 Encoding Fidelity Index (EFI)

Introduced in Paper 8. EFI measures how faithfully an LLM encodes non-English input relative to its English equivalent.

9.1 Theoretical Definition

EFI is defined as the normalised mutual information between the source message X and the model's internal representation $\hat{X}$:

$$\mathrm{EFI} = \frac{I(X;\hat{X})}{H(X)} \tag{19}$$

where $H(X)$ is the Shannon entropy of the source. EFI $= 1$ indicates perfect encoding fidelity; EFI $= 0$ indicates complete encoding failure.

Since $I(X;\hat{X})$ requires access to model internals, a practical proxy is used.

9.2 Proxy Definition

$$\mathrm{EFI}_{\mathrm{proxy}} = \cos\big(\mathbf{e}(\text{non-English input}), \mathbf{e}(\text{English equivalent})\big) \tag{20}$$

where $\mathbf{e}(\cdot)$ is a sentence embedding. English self-similarity serves as the reference: $\mathrm{EFI}_{\mathrm{proxy}}(\text{English}) = 1.0$ by construction.

9.3 Measurement Instrument Dependence

Paper 9 established that $\mathrm{EFI}_{\mathrm{proxy}}$ is substantially instrument-dependent. Under the primary embedding (MiniLM), Kannada EFI $= 0.081$ — a 92% degradation. Under LaBSE (language-agnostic), Kannada EFI $= 0.853$ — only 15% below the English reference. The phenomenon is real under all instruments; its measured magnitude is conservative under English-distilled models.

Language	EFI (MiniLM)	EFI (LaBSE)
English	1.000	1.000
Kannada	0.081	0.853
Tamil	0.073	0.857
Hindi	0.041	0.861

Table 3: EFI by language and embedding model (from Papers 8 and 9).

9.4 Relationship to Var_Ratio

EFI and Var_Ratio are statistically independent under LaBSE ($r = -0.18$, $p = 0.73$, $N = 6$; Paper 9). They measure distinct phenomena: EFI measures encoding quality at the input stage; Var_Ratio measures output consistency at the generation stage. Both are components of the MCH safety framework but require separate evaluation.

10 Statistical Tests

10.1 Domain Separation

Pairwise domain separation is tested using the Mann-Whitney U statistic (non-parametric, two-sided). Effect sizes reported as Cohen's d:

$$d = \frac{\bar{K}^{(d_1)} - \bar{K}^{(d_2)}}{\sqrt{\left(\mathrm{SD}^{(d_1)2} + \mathrm{SD}^{(d_2)2}\right)/2}} \tag{21}$$

10.2 Entanglement Significance

Pearson r tested against $H_0 : r = 0$ using the t-distribution with $df = 28$ (two-sided). Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

10.3 Bootstrap Confidence Intervals (from Paper 9)

For Var_Ratio confidence intervals: 10,000 bootstrap iterations, resampling trials with replacement. 95% CI lower bound exceeding 1.0 indicates significant variance amplification.

11 Embedding Model Specifications

Model	Dim	Category	Role in Paper 6
all-MiniLM-L6-v2	384	EN-distilled	Primary embedding
all-mpnet-base-v2	768	EN-distilled	Robustness check 1
LaBSE	768	Language-agnostic	Robustness check 2

Table 4: Embedding models used for K computation. LaBSE validated as semantically faithful in Paper 9 (non-parallel control: $\Delta = 0.58$, $p = 0.004$).

12 Relationship Between Metrics

The following relationships hold by construction or empirical observation:

$$K = \Delta\mathrm{RCI} \times \mathrm{Var\_Ratio} \tag{22}$$

$$\mathrm{VRI} = 1 - \mathrm{Var\_Ratio} \tag{23}$$

$$\Delta\mathrm{RCI} = \Delta\mathrm{RCI}_{\mathrm{content}} + \Delta\mathrm{RCI}_{\mathrm{order}} \tag{24}$$

$$r_{\mathrm{ent}} \approx +0.76 \text{ in Medical domain (Paper 4)} \tag{25}$$

The conservation constraint states that K is approximately constant within a domain across model architectures:

$$K_i^{(d)} \approx \bar{K}^{(d)} \quad \forall i \in \{1, \ldots, N_d\} \tag{26}$$

with $\mathrm{CV} = 0.162$–0.214 under MiniLM, indicating approximate rather than exact conservation.

13 Data and Code Availability

All raw data, analysis scripts, and computed metrics are available at:

- GitHub: `https://github.com/LaxmanNandi/MCH-Research`
- OSF pre-registration: `https://osf.io/dp8nj/`

Key scripts:

- `paper6_conservation.py` — K computation across all domains
- `paper6_robustness_reembed.py` — Three-embedding robustness
- `paper6_entanglement.py` — Entanglement analysis
- `compile_paper6_data.py` — Final data compilation to JSON

References

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